

---

---

# Project List 2

## 33-467: Astrophysics of the Stars and Galaxy

Fall 2023

---

---

|                                                                    |          |
|--------------------------------------------------------------------|----------|
| <b>1 Binary evolution with COSMIC</b>                              | <b>1</b> |
| 1.1 Orbital Evolution of Binaries in Roche-lobe Overflow . . . . . | 1        |
| 1.2 The Wild World of Binary-Star Evolution . . . . .              | 1        |
| <b>2 Finding Dark Matter with Stellar Streams</b>                  | <b>2</b> |
| <b>3 Black hole kicks with Gaia BH1 and Gaia BH2</b>               | <b>2</b> |
| <b>4 The Fastest Stars in the Galaxy</b>                           | <b>2</b> |
| <b>5 Gravitational Wave Data Analysis</b>                          | <b>2</b> |
| <b>6 How old are those stars ... dynamically speaking?</b>         | <b>3</b> |
| <b>7 Exoplanet Orbital Dynamics with Rebound</b>                   | <b>3</b> |

## 1 Binary evolution with COSMIC

### 1.1 Orbital Evolution of Binaries in Roche-lobe Overflow

This project has both analytic and computational components. In class, we will cover how a binary star's orbit responds to mass exchange through Roche-lobe overflow from one star to its companion. Your first task is to reproduce the derivation under the assumption that the mass transfer is fully conservative. In the Universe, though, it is not required that mass transfer remain fully conservative. In fact, the amount of mass a star can accrete through Roche-lobe overflow is an open research question!

Since we don't know the answer for how conservative mass transfer can be, we parameterize this uncertainty and calculate how different assumptions for how much mass is accreted affect the future evolution of the binary. You should introduce a tuning parameter (often called  $\beta$ ) that represents how much mass is accreted. Before running any code, write down some predictions for how different values for the accretion fraction will alter the future evolution of a binary. Once you have your predictions, install COSMIC and explore the [COSMIC documentation](#) and [COSMIC code paper](#). Next, initialize a binary that will undergo stable mass transfer through Roche-lobe overflow (this will require some guess and check). Using different assumptions for the accretion fraction ('f\_acc'), determine whether your predictions were correct.

### 1.2 The Wild World of Binary-Star Evolution

There are *many* different uncertainties associated with binary evolution due to different interactions between binary components that can occur based on the initial properties of the system and assumptions made for how each interaction should be treated. This means that there are *a lot* of opportunities for exploration. You should explore the [COSMIC documentation](#) and [COSMIC code paper](#) for a summary of the different assumptions made in the code as well as the [Binary Evolution in a Nutshell](#) notes to find an aspect of binary star interactions that you find most interesting. Maybe

it's studying how different assumptions for wind mass loss affects binary populations? Maybe it's investigating how compact object populations depend on the strength of supernova natal kicks? Maybe it's something else!

Whatever topic you choose should not be exactly the same as any other projects, so you may need to coordinate with your colleagues. Once you have a topic to study, you should determine which assumptions are set by flags in COSMIC that affect that topic as well as determine which binary populations will show the largest changes by using different assumptions. You should then summarize how you expect each assumption you make to alter the binary population from its initial configuration at ZAMS. Finally, you should evolve large populations of binaries with COSMIC using the flags you've chosen and make figures which illustrate how each assumption impacts the population.

## 2 Finding Dark Matter with Stellar Streams

One proposed form of dark matter is the  $\Lambda$  Cold Dark Matter ( $\Lambda$ CDM) model which is expected to result in small-scale dark matter halos orbiting along with stars in galaxies. Stellar streams, made up of tidally disrupted globular clusters and dwarf galaxies, are *amazing* tracers of small-scale gravitational overdensities like dark matter clumps.

In this project, you will follow along with a [Data Carpentry Tutorial](#) to reproduce the results of [Price-Whelan et al. 2019](#) which shows that stars in the stellar stream Gaia GD-1 may have been perturbed due to a dark matter clump. This project will involve building skills in astroquery, joining different data sets, and selecting regions of interest in a dataset. By the end of the project, you should have figures which show the Gaia GD-1 stream and illustrate why the authors argue that the stream has been perturbed by a small-scale dense object that could be a dark matter subhalo. As part of this project, you should also summarize what stellar streams are, what the  $\Lambda$ CDM model predicts and why it is favored by the scientific community, and how future observations could aid in discoveries of dark matter clumps.

## 3 Black hole kicks with Gaia BH1 and Gaia BH2

The recent discoveries of [Gaia BH1](#) and [Gaia BH2](#) were some of the most exciting discoveries in 2023 (at least according to Prof. Breivik). Since there is no accretion in either binary, the last thing to alter the orbit of the binary and its trajectory in the Galaxy was the natal kick that may or may not have been imparted during the formation of each black hole. This means that the current orbit of each binary in the Galaxy places a pretty strong constraint on the black hole kicks!

For this project, you should gather the observed properties of each binary from their respective papers. You should then download and install [COSMIC](#) and check out the documentation. In particular, you should find the section that shows how to restart a binary and gives a specific example for Gaia BH1. Once you are able to produce a population of black hole binaries, you should explore the degeneracies between the pre-explosion separation, the strength and direction of the black hole natal kick, and the final orbital period and eccentricity. Finally, you should repeat the exercise with Gaia BH2's parameters. Do Gaia BH1 and Gaia BH2 have similar kick strengths?

## 4 The Fastest Stars in the Galaxy

Type Ia supernovae are extremely important in both astronomy and cosmology because they are *standardizable* candles. This means that with some calibration, their brightness can be used to infer their distance. This technique was famously applied to measure the expansion of the Universe which was found to be accelerating rather than just expanding. This finding led to Saul Perlmutter, Brian Schmidt, and Adam Riess being awarded the Nobel Prize in 2011.

Today, one of the main concerns about Type Ia supernovae is whether they are caused from the mergers of two white dwarfs (double degenerate) or due to accretion onto a white dwarf (single degenerate) or some combination of both! One of the favored channels is the D6 scenario whose name is based on the dynamically-driven double detonation of double degenerate (6 d's!) binaries. For this project, you should first do a literature search to understand what the D6 scenario is and what observable signatures a D6 source would lead to. [Shen+2019](#) is a good starting place, but there are lots of good references in Section 2 of the paper. Once you have a handle on D6, use Gaia DR3 to find D6 candidates based on the observable signatures of the channel. By the end of this project, you should have a list of candidates and be able to argue why you think they are or are not originating from the D6 channel.

## 5 Gravitational Wave Data Analysis

The discovery of Gravitational Waves by the LIGO-Virgo Collaboration in 2015 marked a feat of science nearly 100 years in the making following their prediction by Einstein's theory of General Relativity. The detection of two merging black holes that were more than a billion light years away led to the 2016 Nobel Prize being awarded to Rai Weiss, Kip Thorne, and Barry Barish. The work was carried out by *over a thousand* people though – the detection was a physics experiment in and of itself!

Luckily for us, the LIGO Science Collaboration has produced tutorials for interested folks to download and play with their data so that you can learn exactly how gravitational wave data analysis works. For this project, you should work through the "Guide to GW detections and noise" tutorial on the [GW open science center website](#). Once you've finished

that tutorial, you can move onto the "Searching for astrophysical sources" section to find an inspiring source. By the end of the project, you should have a lot of figures and be able to describe how each analysis step works.

## 6 How old are those stars ... dynamically speaking?

It is often *very* hard to tell whether a star is very old based on photometry alone since main sequence lifetimes for your average star are several billion years. One way that we can measure the age is through kinematics because old, low-metallicity stars are empirically known to orbit the Galaxy at larger vertical distances from the center of the disk than younger, higher-metallicity stars. In this project, you will explore how well kinematic ages approximate the true age and metallicity of stars.

First, you should download [Gala](#) and work through the tutorial which shows how to integrate orbits of stars observed by Gaia. Once you have worked through the tutorial, you should gather a set of stars with known metallicities and integrate their orbits. Do the orbits look like you'd expect based on the metallicity? If not, what are some reasons that there could be discrepancies. Based on the current position of the stars, is it safe to only use the position as an age indicator? How much information does the addition of the proper motion add?

## 7 Exoplanet Orbital Dynamics with Rebound

The discovery of over 5,000 exoplanets has made it clear that our own solar system is not the only configuration for which planets can orbit stars on long-term stable orbits. One of the biggest surprises of the 2010's was that Jupiter-mass planets were found to orbit their host stars closer than Mercury orbits in our own solar system. These planets were named "Hot Jupiters" due to their proximity to their host star and were subsequently the subject of many theoretical papers which tried to determine whether these planets could have been scattered through gravitational interactions with other planets, slowly migrated inward over time, or just been born there in the first place.

For this project you should first do a literature search in ADS to trace the reasoning for different theories and the evidence that backs each of them. You should then download [REBOUND](#), an open-source code that you can use with Python or C and calculates the evolution of gravitationally bound systems extremely accurately. There are several examples for how REBOUND works for planet-star systems which you should work through. After this, you should modify the examples to try to understand how different assumptions for Hot Jupiter formation play out when integrating different initial configurations of exoplanet systems.